

Mathis Pink

PhD Candidate in Computer Science
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RESEARCH

I study memory across artificial and biological systems: how intelligent systems encode experience in context, reinstate that context during remembering, and control how past experience shapes ongoing computation. My work combines machine learning, mechanistic analysis of language models, human behavioural experiments, and computational neuroscience.

POSITIONS

- 2024 – present **PhD Candidate in Computer Science**
Max Planck Institute for Software Systems (MPI-SWS) & Saarland University, Saarbrücken
Advised by Mariya Toneva and Isabel Valera. Member of the Research Training Group *Neuroexplicit Models of Language, Vision, and Action* (DFG GRK 2853/1). Expected completion 2028.
- Mar – Dec 2023 **Research Assistant**
University of Osnabrück, Germany
With Adrien Doerig and Tim Kietzmann. Deep reinforcement learning, visual question answering, foveal image sampling, saccadic remapping, and recurrent convolutional networks.

EDUCATION

- 2020 – 2023 **M.Sc. Cognitive Science**
University of Osnabrück, Germany
Specialisation: Neuroinformatics & Artificial Intelligence. Final grade **1.0** (German scale: 1.0 highest, 4.0 pass).
Thesis: *Learning Where to Look in Natural Visual Scenes*, advised by Tim Kietzmann and Adrien Doerig.
- 2015 – 2020 **B.A. Philosophy & Economics**
University of Bremen, Germany
Specialisation: theoretical philosophy of mind. Final grade **1.4**.
Thesis: *Phenomenal Consciousness as Affective Meta-Representation*.

PUBLICATIONS

- 2026 **Temporal Context Reinstatement Drives Episodic-Like Order Memory in Long-Context Language Models**
Mathis Pink, Vy A. Vo, Qinyuan Wu, Jianing Mu, Javier S. Turek, Uri Hasson, Kenneth A. Norman, Sebastian Michelmann, Alexander Huth, Mariya Toneva
International Conference on Machine Learning (ICML), 2026. arXiv:2607.22575 · OpenReview · code
- 2026 **Tug-of-war between idioms' figurative and literal interpretations in LLMs**
Soyoung Oh, Xinting Huang, **Mathis Pink**, Michael Hahn, Vera Demberg
Conference of the European Chapter of the Association for Computational Linguistics (EACL), long paper, 2026. doi:10.18653/v1/2026.eacl-long.135
- 2025 **Choice history biases in dyadic decision making**
Ann Huang, **Mathis Pink**, Viktoria Zemliak, Artur Czeszumski, Peter König
Scientific Reports 15, 11420, 2025. doi:10.1038/s41598-025-96182-5
- 2025 **Position: Episodic Memory is the Missing Piece for Long-Term LLM Agents**
Mathis Pink, Qinyuan Wu, Vy A. Vo, Javier Turek, Jianing Mu, Alexander Huth, Mariya Toneva
Preprint, 2025. arXiv:2502.06975
- 2024 **Assessing Episodic Memory in LLMs with Sequence Order Recall Tasks**
Mathis Pink, Vy A. Vo, Qinyuan Wu, Jianing Mu, Javier S. Turek, Uri Hasson, Kenneth A. Norman, Sebastian Michelmann, Alexander Huth, Mariya Toneva
Preprint, 2024. arXiv:2410.08133

PRESENTATIONS

July 2026	Temporal Context Reinstatement Drives Episodic-Like Order Memory in Long-Context Language Models Poster, 43rd International Conference on Machine Learning (ICML), Seoul, South Korea.
Aug 2025	Evaluating Models of Naturalistic Episodic Memory Poster, 8th Conference on Cognitive Computational Neuroscience (CCN), University of Amsterdam, Netherlands. With Shashwat Saxena and Mariya Toneva.
Sept 2024	Assessing Episodic Memory in LLMs with Sequence Order Recall Tasks Poster, NEAT (Neuro-AI-Talks), Osnabrück University, Germany.

DATASETS AND SOFTWARE

2026	Temporal Context Reinstatement Code and data for mechanistic analysis of temporal context reinstatement in long-context language models. github.com/mathispink/temporal-context-reinstatement
2024	SORT and Book-SORT Sequence Order Recall Tasks: an evaluation paradigm and naturalistic long-term memory dataset for measuring episodic-like memory in language models, validated against a human experiment. github.com/bridge-ai-neuro/SORT
in development	ICMemory-Kit Causal interventions under KV-cache reuse for long-context memory experiments, enabling efficient mechanistic analysis on limited compute. Pre-release. github.com/mathispink/icmemory-kit

TEACHING

2025	NeuroAI – Neuroscience and AI (summer school) Teaching Assistant, CAJAL Advanced Neuroscience Training Programme, Champalimaud Centre for the Unknown, Lisbon, Portugal. 14 July – 1 August 2025.
2021 – 2023	University of Osnabrück – Teaching Assistant Machine Learning for Cognitive Computational Neuroscience (winter 2022/23); Implementing ANNs with TensorFlow (winter 2021/22 and winter 2022/23, as co-instructor); Deep Reinforcement Learning (summer 2022); Machine Learning (summer 2021, summer 2022); Computer Vision (winter 2021/22).
Aug 2021	Deep Learning (summer school) Teaching Assistant, Neuromatch Academy.

ACADEMIC SERVICE

2024 – 2026	Reviewer – Annual Meeting of the Association for Computational Linguistics (ACL)
2026	Reviewer – Annual Meeting of the Cognitive Science Society (CogSci)

TECHNICAL & METHODOLOGICAL EXPERTISE

Methods	Brain–AI representational alignment (encoding models, representational similarity analysis); mechanistic interpretability (activation patching, causal interventions); design and analysis of human behavioural experiments; Bayesian modelling.
Modelling	Pretraining language and vision models from scratch; finetuning and representational steering; deep reinforcement learning.
Infrastructure	Large-scale LLM inference and serving (vLLM, SGLang); HPC and SLURM cluster workflows; agentic coding.
Frameworks	PyTorch, JAX, TensorFlow, Hugging Face Transformers, NumPyro.
Languages	German (native), English (fluent), French (basic).